Zoya Rahman

US Citizen | rahmanzn2016@gmail.com | She/They | DMV | LinkedIn | portfolio | Github

RESEARCH INTERESTS

Human-Computer Interaction, Computer Vision, 3D Reconstruction, Representation Learning, Computational Imaging

EDUCATION

University of Maryland: College Park, GPA: 3.59 08/2023 – Present
B.S. in Computer Science: Data Science Concentration Expected May 2027
B.A. in Immersive Media Design: Emerging Creatives Track

Computer Science Coursework: Introduction to HCI; Computer Vision; Computational Photography; Design and Analysis of Algorithms; Advanced Data Structures; Linear Algebra; Applied Probability and Statistics; Machine Learning.

Immersive Media Coursework: Computational Virtual Reality; Computational Media; Image + Time; Code & Form (Fabrication); Experiential Computing; 3D Fundamentals; Advanced Digital Media Theory

PUBLICATIONS AND MANUSCRIPTS

Titles generalized and co-authors omitted for confidentiality during review.

Z. Rahman (first author) et al. Deep learning for satellite-based plankton estimation from hyperspectral imagery.
Under review, 2026.

Z. Rahman* et al. Remote Homology Detection using Representation Learning.
Under review, 2026.

Z. Rahman* et al. Structural Representation Learning for Protein Structures.
Under review, 2026.

*Co-first author

RESEARCH EXPERIENCE

Undergraduate Research Assistant 08/2026 – Present
Embodied Dynamics Laboratory: Dr. Jun Nishida University of Maryland
Wearable Perception Study: user study of wearable lenses that selectively filter the wearer's view during performance tasks.

- Redesigned the baseline condition's view-blocking effect, replacing full occlusion with blur based on visual-attention literature (Unity, C#, Meta Quest 3).
- Ran participant study sessions.
- Worked in Autodesk Fusion for device fabrication.

Embodied Skill Study: user study on whether multimodal wearable signals can sense motor chunk formation and lead errors.

- Developing user study setup.
- Conducting pilot study testing.
- Contributing to data analysis and annotation.

Independent Undergraduate Research 09/2026 – Present
Iceberg 3D Reconstruction: Prof. Cy Keener University of Maryland

- Testing whether existing vessel-based camera imagery and sonar data can support 3D reconstruction of floating icebergs, as a feasibility study.
- Preprocessing and aligning the image and sonar data; reproducing a vessel-based underwater photogrammetry approach and extending prior neural radiance field (NeRF) work on Arctic ice reconstruction.

Undergraduate Research Project Leader 01/2026 – 09/2026
Hyperspectral Deep Learning: Dr. Myungin Lee University of Maryland

- Proposed and led an independent project estimating plankton bloom biomass and community composition from NASA PACE hyperspectral satellite imagery and in-situ oceanographic data.
- Designed compact spatiotemporal CNN models for settings where paired satellite and in-situ training examples are scarce.
- Built a grouped evaluation protocol that keeps near-duplicate observations out of both training and test sets.
- Grew the work from an independent study into a team project; first-author manuscript under review.

Undergraduate Research Assistant 09/2025 – 10/2026
Protein Structure Representation Learning: *Dr. Fardina F. Alam* University of Maryland

- Developed hybrid autoencoder-CNN architectures (PyTorch) for protein structure representation and classification on UMD HPC clusters.
- Evaluated whether learned representations generalize to protein families held out from training, using statistical testing.
- Reviewed and presented literature to guide team model design.
- Co-first author on two manuscripts under review.
- Developed the remote homology detection project from scratch as an application of the team's representation learning methods.

Biomedical Machine Learning Intern 11/2024 – 01/2025
U.S. Food and Drug Administration, CDRH / OSEL, Division of Imaging, Diagnostics, and Software Reliability (*DIDSR*) Silver Spring, MD

- Evaluated the credibility of an ML model for intracranial hemorrhage detection in pediatric versus adult CT scans, working under PhD researchers.
- Curated and processed CT datasets; designed and ran 4 experiments with full methodology documentation.
- Analyzed and visualized results (Python, Matplotlib) and briefed findings to the research team.

SELECTED RESEARCH AND TECHNICAL PROJECTS

Memento of Self: Sparse-View Photometric Reconstruction Installation Summer 2026
Project Lead

- Investigated whether appearance-based (photometric) reconstruction degrades more gracefully than geometric accuracy under sparse camera views.
- Built a sparse-view Gaussian Splatting pipeline that reconstructs visitors from a multi-camera array as painterly 3D portraits.
- Designed and fabricated the booth (housing, camera mounts, lighting and trigger systems) with woodworking and physical computing.
- Found that a seven-camera array produced visually convincing reconstructions with deformed geometry, motivating my research on human-aligned reconstruction.

Into the Veins: Interactive Hand-Tracking Installation Spring 2026
Independent Final Project, *ARTT* 370 (*TouchDesigner*)

- Built a real-time interactive animation in TouchDesigner in which particle "cells" with trails follow each of the viewer's hands via MediaPipe hand tracking, with an independent trail system per hand.
- Drove the background animation from the audio amplitude of the score for an audio-reactive environment.
- On continuous display in the Immersive Media Design Department since July 2026.

Telekinesis: Give Me A Hand Spring 2025
Project Lead and Gesture Tracking Developer

- Led a five-person team building a controller-free VR obstacle course driven entirely by hand tracking (Unity, C#, Meta Quest 2).
- Reduced gesture-recognition latency for real-time interaction; ran user testing to refine gesture controls for accessibility.

Analyzing Student Success Factors Summer 2025
Project Lead: *Data Science CMSC* 320

- Led a four-person team; developed the ML model and statistical analysis predicting student success from a real course roster dataset.
- Published the workflow as a public step-by-step tutorial.

TEACHING AND MENTORSHIP

Undergraduate Teaching Assistant Spring 2026
IMDM 290: *Collaborative Studio* 1: *Image + Time* University of Maryland

- Supported a capstone course built on Unity (C#) and collaborative GitHub workflows through project development.
- Held office hours four days a week for debugging and GitHub support.
- Supported a 28-student section; met with six project groups twice weekly.
- Documented lecture notes for absent students.
- Led lecture during the professor's absence.

Immersive Media Design Undergraduate Administrative Services 07/2026 – Present
Peer Advisor University of Maryland

- Advised approximately 30 current and prospective students in my first three months through drop-in advising and academic planning, including four-year plans, semester scheduling, and course selection.
- Serve as the student point of contact during departmental transitions and staffing gaps.
- Participated in interviewing candidates for new administrative staff.
- Support department outreach, including open houses and tabling events.

LEADERSHIP & SERVICE

Co-President, IMMERSE 09/2025 – Present
Immersive Media Makers Exploring Reality, Simulation, and Experience University of Maryland

- Led the transition of the IMD student advisory board into an independent student organization for art and technology.
- Wrote its constitution and executive and committee structure.
- Secured $1,000 in departmental funding for launching flagship events.
- Designed organization website and brand.
- Grew a community of 100+ students across our channels, with average event attendance of 40+, including students outside the major.

Co-Director of Design, Association for Women in Computing 06/2025 – Present
Creating AWC's digital presence & brand identity. University of Maryland

- Lead AWC's visual identity, website UI/UX, and front-end development.
- Manage and mentor the design team: task delegation, regular check-ins, and liaison to the executive board.

PRESENTATIONS, EXHIBITIONS, AND HONORS

Student Undergraduate Research Conference: Poster Presentation
Z. Rahman, A.R. Ariff, M. Lee. "*PACE OCI Remote-Sensing Reflectance for Phytoplankton Community Classification and Short-Horizon Bloom-Risk Estimation*." University of Maryland Student Undergraduate Research Conference (SURC), July 2026. Presented by Z. Rahman and A.R. Ariff.

Exhibitions

- *Memento of Self*:
  - New Works Incubator Showcase, University of Maryland, Summer 2026.
  - Henson Mania: A Night of Art, Innovation & Imagination, University of Maryland, Summer 2026.
- *Into the Veins*, ongoing display, Immersive Media Design Department, University of Maryland, July 2026 – Present.
- *Telekinesis*: *Give Me A Hand*, Immersive Media Design Showcase, University of Maryland, Spring 2025.

Honors
Dean's List, College of Computer, Math, and Natural Sciences & College of Arts and Humanities, University of Maryland.

- Spring 2026, Fall 2025, Spring 2025

Selected Student Speaker, Immersive Media Design Senior Capstone Exhibition, University of Maryland, Spring 2026.

- Chosen from the junior class to speak at the senior capstone exhibition based on demonstration of departmental leadership.

SKILLS

- **Programming:** Python, Java, C, C#, HTML/CSS, Git
- **Machine learning:** PyTorch, CNNs, autoencoders, spatiotemporal models, statistical evaluation, HPC clusters
- **Vision and 3D:** Gaussian Splatting, NeRF, MediaPipe, photogrammetry, depth sensing, hyperspectral imagery, CT imaging
- **XR and fabrication:** Unity, Unreal Engine, Meta Quest 2 & 3, hand tracking, TouchDesigner, Blender, SolidWorks, CNC fabrication, physical computing
- **Research methods:** Human-subjects studies, pilot testing, eye tracking, physiological sensing (heart-rate variability, skin conductance), user testing
- **Design:** Figma, Adobe Illustrator, Procreate, Canva